

05.4 Architecture Methodology

How 210 framework tools were selected, pruned, and operationalised into context-specific Implementation Architectures for the UK and Netherlands

Companion to: 05.1 UK Implementation Architecture, 05.2 NL Implementation Architecture, 05.3 Architecture Codebook

1. Overview

1.1 Purpose

This document specifies the methodology of Stage 3 of the A Safe Bet translational design programme: how the 210-tool framework produced in Stage 2 was selected from, pruned, and operationalised into two context-specific Implementation Architectures: one for the UK (SunVegas, regulated under the Licence Conditions and Codes of Practice) and one for the Netherlands (Holland Casino Online, regulated under the Dutch Remote Gambling Act, KOA 2021).

Both architectures are limited to online casino games on the two platforms; sports betting, poker, lottery and land-based products are out of scope, and tools whose mechanism applies only to those products were not carried into the architectures.

The methodology covers the selection logic by which framework tools entered the architecture; the variant generation process by which selected tools were operationalised at multiple risk tiers and configurations; the policy-rule authoring approach; the exclusion taxonomy; and the cross-jurisdictional design decisions that produced the two architecture variants.

1.2 Where Stage 3 sits in the design programme

Stage 3 is the deployment-specification stage of a three-stage translational design programme. Stage 1 (the systematic review and taxonomy) produced 410 tool records across a 45-field extraction codebook. Stage 2 (the RG Intervention Framework, file 04.1) consolidated those records into 210 framework tools structured along six categorical dimensions plus Taxonomy Source. Stage 3 selects from the framework, prunes by regulatory and operator constraints, and operationalises selected tools as deployable variants with full configuration parameters and player-facing copy.

The translational design programme runs from research evidence (Stage 1) to consolidated framework (Stage 2) to deployable architecture (Stage 3). Stage 3 outputs are the deployable configurations of the A Safe Bet framework: two context-specific Implementation Architectures.

1.3 Outputs

- 05.1 UK Implementation Architecture: 14 architecture tools, 94 variants, 78 policy rules across 10 sections, 192 message templates
- 05.2 NL Implementation Architecture: 13 architecture tools, 87 variants, 133 policy rules across 12 sections, 215 message templates
- 05.3 Architecture Codebook: column definitions, controlled vocabularies, decision rules
- 05.4 Architecture Methodology: this document

2. The translational design approach

2.1 Risk tier as the primary unit of design

The architectures are organised primarily around player risk tier. Three tiers are operationalised (R1 low risk, R2 medium risk, R3 high risk), with Young Adult (YA) treated as an independent overlay

applied on top of the standard tier configurations. Every tool that enters the architecture is operationalised separately at each tier, and tier-specific tool variants are connected by a coordinated tier-based delivery logic that tightens the same tool at higher risk levels and relaxes it at lower levels.

The decision to organise around risk tier rather than around tool was a deliberate methodological choice. The framework's Target Population dimension provides the conceptual scaffolding for risk-based stratification; Stage 3 operationalises that scaffolding by treating the tier as the primary axis of design and the tool as the secondary axis. This reverses the framework's natural reading order (tools first, target populations second) but produces a deployment artefact that matches how operators configure their RG systems: by player segment first, by tool feature second.

Each risk tier carries its own messaging tone of voice, parameter values, and assignment rules. R1 messaging is built around positive play framing, gambling literacy, and educational content. R2 messaging adopts motivational interviewing principles and psychotherapeutic reframing. R3 messaging is declarative, non-stigmatising, and tool-specific: it refers the player back to the tool's purpose rather than persuading them. The tone progression operationalises a synthesis of framework content concepts; mapping is documented in 04.3 (Risk-tier messaging tone of voice).

2.2 Adaptive parameter tailoring

The architectures implement an adaptive design principle: the same tool becomes more restrictive at higher risk tiers and less restrictive at lower tiers, with multiple parameters tightening or loosening together. This is the principal contribution of Stage 3: an evidence-based, adaptive online RG system.

Concretely, six adaptive mechanisms operate in parallel across tiers:

- **Threshold tightening:** the same threshold percentage (typically 0.8) applies across tiers, but tier-specific thresholds fire on different bases (player-set caps at R1, system-derived caps at R3).
- **Limit-set mode shift:** as risk increases, control of the limit value shifts from player to system. UK uses 10 distinct limit-set modes ranging from `player_defined` through `system_1.5x` to `system_enforced`; NL uses 3 modes.
- **Asymmetric cooldown:** increase requests are gated by tier-specific cooldown values; decreases apply immediately. UK applies six different cooldown values across tiers (24h, 48h, none, plus disable and cannot-disable variants); NL uniformly applies 168h.
- **Content type collapse:** PPF and EPF differentiation at R1 and R2 collapses to a single 'unified' content type at R3, reflecting an operational simplification at high-risk tier.
- **Frequency caps tightening:** increase request frequency is constrained more tightly at higher tiers.
- **Tool toggle by tier:** some tools activate only above R1 (e.g., Bonus Ban and Promo Ban in the UK architecture).

This adaptive logic is not a single framework tool but a synthesis of multiple framework dimensions: Target Population provides the tier scaffolding, the Type sub-vocabulary of Trigger provides threshold mechanics, Personalisation provides the user-system tailoring axis, and the Reversibility sub-vocabulary of Governance provides the asymmetric increase-decrease principle. The synthesis is documented in 04.3 (Adaptive parameter tailoring across risk tiers).

2.3 Player autonomy as a design principle

The architectures preserve player autonomy within the constraints of operator UI capabilities. Two interaction principles inherit from the framework: personalised data interpolation and choice-response design.

Personalised data interpolation operationalises the framework's FEED-PERSONAL family (T085 Personalised Normative Comparison, T086 Self-Estimation vs Actual, T089 Personalised PGSI Feedback) and the MSG-POP tools that interpolate behavioural data (T060 Time-Elapsed, T061 Spend-Elapsed, T067 Loss-at-Threshold). PPF and EPF content types interpolate real player data: a player viewing 'You have spent £127 this week' is in the same intervention category as the framework's T061.

Choice-response design preserves player autonomy within messaging surfaces. Where the framework's T069 (Forced-Choice Continuation) and T068 (Pre-Session Goal-Setting) provide choice as a tool-level mechanism, the architecture extends this principle across all messaging surfaces by avoiding single-action dismissals. Each message offers the player meaningful interaction options (e.g., 'Review My Limits | Continue Play | Exit') rather than a single dismissive close. Choice mechanics are bounded by what the operator UI can support and by what regulation permits.

3. The selection process

The 210 framework tools were filtered through four sequential criteria to identify the deployable subset for each architecture. The criteria are applied in order, with each step pruning candidates ineligible at that stage. The same procedure is applied independently to the UK and NL architectures, producing two parallel selections.

3.1 Filter 1: Regulatory necessity

Each candidate tool was first evaluated against the jurisdiction's regulatory framework. Tools mandated by regulation entered the architecture as Mandatory (M) regardless of other criteria. Tools prohibited or already covered by superseding regulation were excluded under the regulatory exclusion class (e.g., Bonus Ban and Promo Ban in NL, where KOA 2021 already restricts bonuses).

Regulatory necessity in UK derives from the Licence Conditions and Codes of Practice (LCCP); in NL from KOA 2021 plus operator voluntary commitments documented in HC_voluntary_2022. The source citations in the policy_rules sheet trace each rule back to its regulatory provenance.

3.2 Filter 2: Evidence prioritisation

Tools surviving the regulatory filter were prioritised by Evidence Status. Empirically tested tools (those with direct evaluation in the systematic review evidence base) were preferred. Within the Empirically tested tier, tools backed by the convergent evidence set (38 records from 35 papers from Stage 1 representing the highest-confidence findings) were prioritised further.

Of the 58 unique framework tools that entered the architectures, 41 (70.7%) are Empirically tested and 21 (36.2%) have at least one convergent-evidence-set paper in their underlying provenance. This compares to a framework baseline of 22.4% convergent-set membership, demonstrating that the architectures preferentially selected tools backed by the strongest empirical evidence. Among the 41 Empirically tested tools the architectures draw on, 19 (46.3%) have convergent evidence: above the framework baseline rate of 30.8%.

Quantitative selection demonstration is documented in 04.3 (Evidence-based selection).

3.3 Filter 3: Assignment status

Tools surviving the evidence filter were assigned a status indicating their relationship to player onboarding: Mandatory (M) tools are assigned to all players at registration; Optional (O) tools are available as supplementary configuration. Mandatory Overlay (M_overlay) tools are assigned to specific player segments (e.g., Young Adults) on top of their standard tier configuration.

Assignment status was determined by combining regulatory necessity (Filter 1) with operator policy and the framework's Governance.Control dimension. UK distribution: 26 variants Mandatory (27.7%), 62 Optional (66.0%), 6 Mandatory Overlay (6.4%). NL distribution: 26 variants Mandatory (29.9%), 59 Optional (67.8%), 2 Mandatory Overlay (2.3%).

3.4 Filter 4: Operator and technical constraints

The final filter accounted for operator-specific constraints: UI capabilities, integration timeline, technical infrastructure. Tools failing this filter were excluded under the operator or technical exclusion class (e.g., Automatic Withdrawal Limit excluded from UK architecture because UK banking infrastructure does not support the required automatic-debit operation).

Tools surviving all four filters entered the architecture and were operationalised in Section 4.

3.5 Resulting tool inventory

UK architecture: 14 architecture tools (Deposit Limit, Loss Limit, Bet Limit, Login/Playtime Limit, Deposit Count Limit, Bonus Ban, Promo Ban, Scheduled Break, Reality Check, Session Cooldown, RG Homepage, Cheat Sheet, Positive Play Self-Assessment, Dark Flow Self-Assessment).

NL architecture: 13 architecture tools, the same as UK except: Bonus Ban and Promo Ban excluded (regulatory class); Automatic Withdrawal Limit added (regulatory class, KOA 2021 mandates a wallet balance limit).

12 tools are deployed in both architectures; 2 are UK-only (Bonus Ban, Promo Ban); 1 is NL-only (AWL).

4. Variant generation

Each tool selected for an architecture was operationalised as multiple variants. A variant is a tool-tier-configuration combination: the same tool deployed at a different risk tier or with a different content type, parameter value, or limit-set mode produces a distinct variant.

4.1 Variant generation logic

Variants are generated by stratifying each tool along three primary axes: risk tier, content type, and configuration parameter. The axes interact such that not every combination produces a variant; design decisions at one axis can collapse variation at another. Examples:

- At R3, the PPF/EPF content type distinction collapses into a single 'unified' content type, producing one R3 variant where R1 and R2 each have two (PPF + EPF).
- Where a tool's parameter is regulator-mandated (e.g., NL increase cooldown of 168h across all tiers), variant counts decrease because there is no room for tier-based parameter variation.
- Where a tool offers multiple limit-set modes (e.g., UK Session Cooldown with player_elected_30_60CD, player_elected_30_60_90CD, system_1.5x_daily), each mode produces its own variant.

This logic produces UK 94 variants from 14 tools (mean 6.7 per tool, range 5–12) and NL 87 variants from 13 tools (mean 6.7 per tool, range 5–10). Per-tool variant counts are documented in the architecture files (05.1 and 05.2).

4.2 Variant parity across architectures

Of the 12 tools deployed in both architectures, 10 have identical variant counts in UK and NL. Two diverge: Loss Limit (UK 12, NL 8) and Login/Playtime Limit (UK 6, NL 8). The Loss Limit divergence arises from UK using two YA sub-age-band variants (YA_18 and YA_21) where NL uses

a single YA tier; the Login/Playtime divergence arises from NL’s mandatory regulatory framing requiring more variants to specify across tier transitions.

4.3 Variant ID format

Each variant has a unique identifier in the form [JUR].PA.[tier].[M/O].[category.tool#].[variant#], specified in 05.3 Architecture Codebook Section 2. Variant IDs are stable: once assigned, an ID is never reused or reassigned. Excluded variants retain their assigned IDs in the excluded_tools sheet.

4.4 Specification status

Each variant is marked fully_specified (reported in the main manuscript as operationally complete) or specified_not_built (specified but not built). Variants marked fully_specified have complete operational specification (design parameters set, message copy developed where applicable, policy rules in place). Variants marked specified_not_built have complete specification but were not built within the project timeline, typically because the tool is in the architecture’s design but its production deployment was deferred.

UK distribution: 69 fully_specified (73.4%), 25 specified_not_built (26.6%). NL distribution: 72 fully_specified (82.8%), 15 specified_not_built (17.2%). The higher proportion of fully_specified variants in NL reflects KOA 2021’s narrower set of permitted parameter values, which reduces the design space and accelerates specification work.

5. Policy rule derivation

Each architecture is governed by a set of policy rules that operationalise system-level behaviour: tool assignment logic, threshold trigger conditions, limit-set mode rules, cooldown specifications, tailoring formulae, refresh timing, and Young Adult overlay rules. UK has 78 rules across 10 sections; NL has 133 rules across 12 sections.

5.1 Rule sources

Each policy rule is grounded in a documented source. The source taxonomy distinguishes regulatory authority, operator specification, empirical data, and combinations:

Data	Empirical operator data: typically observed player behaviour distributions used to set adaptive parameter values (UK only).
IMS_spec	Operator Internal Management System specification: the operator’s own implementation specification.
IMS + Data	Combination of IMS specification and empirical data (UK only).
KOA_2021	Dutch Remote Gambling Act 2021: primary regulatory source (NL only).
KOA_2021 + IMS_spec	KOA grounding plus operator implementation specification (NL only).
LCCP	UK Licence Conditions and Codes of Practice: primary UK regulatory source (UK only).
HC_voluntary_2022	Holland Casino voluntary commitment documented in 2022 (NL only).
refresh_calendar	Refresh event calendar reference: common to both architectures.

5.2 UK source distribution (78 rules)

- IMS_spec: 35 rules (44.9%): operator-driven specification.
- Data: 36 rules (46.2%): empirical-data-driven.
- IMS + Data: 5 rules (6.4%): combination.

- LCCP: 1 rule (1.3%): direct regulatory citation.
- refresh_calendar: 1 rule (1.3%).

5.3 NL source distribution (133 rules)

- IMS_spec: 101 rules (75.9%): operator-driven specification.
- KOA_2021 + IMS_spec: 21 rules (15.8%): combination.
- KOA_2021: 9 rules (6.8%): direct regulatory citation.
- HC_voluntary_2022: 1 rule (0.8%): operator voluntary commitment.
- refresh_calendar: 1 rule (0.8%).

5.4 Cross-jurisdictional regulatory grounding

The source distributions reveal a substantive contrast between the two architectures. UK rules are predominantly grounded in operator data and IMS specification, with regulatory citation (LCCP) appearing only once. NL rules are predominantly IMS_spec, but with significant direct regulatory citation (KOA_2021 alone or combined): 30 of 133 NL rules (22.6%) are directly grounded in KOA 2021, where only 1 of 78 UK rules cites LCCP directly.

This pattern frames a substantive methodological point: NL parameter uniformity is regulatory (KOA 2021 prescribes specific parameter values that propagate through the architecture as direct regulatory citations), while UK parameter variability is operator-driven (LCCP permits operator design discretion, and most UK rules are grounded in operator data and specification rather than in regulatory text).

5.5 Rule confirmation status

Each rule is marked CONFIRMED or NEEDS_CONFIRM. CONFIRMED rules have been signed off by the operator's compliance and product teams. NEEDS_CONFIRM rules remain pending operator confirmation; the rule's draft value is recorded but should be treated as provisional pending sign-off.

UK: 78 of 78 rules CONFIRMED (100%). NL: 131 of 133 CONFIRMED (98.5%) plus 2 NEEDS_CONFIRM (one in Login/Playtime Tailoring Formula, one in Young Adult Rules). The two NEEDS_CONFIRM rules represent open items pending operator sign-off at the time of architecture capture; they are reported here as caveats rather than gaps.

5.6 Section structure differences

UK uses 10 policy rule sections; NL uses 12. The additional NL sections decompose tailored limit types into separate tailoring-formula sections: NL has dedicated sections for Login/Playtime Tailoring Formula (Section 6), AWL Tailoring Formula (Section 7), and Deposit Count Limit Tailoring Formula (Section 10), where the UK architecture collapses these into shared rules within Sections 5–7. The NL section count reflects KOA 2021's more detailed regulatory requirements; the UK section count reflects operator design discretion under LCCP.

6. Message template development

Player-facing copy was developed for each variant marked fully_specified that requires player-facing messaging. UK has 192 message templates across 10 tools; NL has 215 templates across 11 tools. Mean word count is 63.9 (UK) and 61.4 (NL); both architectures use a 29–115 word range.

6.1 Development approach

Message templates were developed from the framework's MSG-POP, MSG-PUSH, MSG-STATIC, and FEED-PERSONAL families, filtered for online In-product delivery and Empirically tested or convergent-set evidence status. Each template was authored to fit a specific delivery point in the player flow, identified by message_stage and message_surface.

Copy authoring was iterative. An initial draft drew on framework tool descriptions and the messaging tone-of-voice protocol for the variant's risk tier (R1 positive play; R2 motivational interviewing; R3 declarative). Drafts were then reviewed against three criteria: (a) tier-appropriate tone; (b) clear action options for the player (choice-response design); (c) operator UI feasibility within the available message surface (modal popup, toast, static page, or error message).

6.2 Stage parallelism

Both architectures use 19 distinct message stages. Seven stages appear in both architectures with consistent semantics: breach_100_popup, decrease_toaster, end_confirmation_popup, increase_request_popup, pending_increase_popup, threshold_25_popup, threshold_80_popup. Six stages are UK-only (relating to break management and login reminders); six are NL-only (relating to break activation and reality-check pop-ups).

Stage naming differs in some cases between architectures. For example, UK uses break_initiation_popup_onetime where NL uses break_initiation_one_time_popup. The underlying stages are functionally equivalent; the divergence is operational naming rather than design-driven.

6.3 Personalisation in messaging

PPF and EPF are defined in 05.3 Section 3.1.6; this section describes their methodological role and the R3 unified-content collapse.

PPF (Personalised Player Feedback) and EPF (Enhanced Player Feedback) content types interpolate player-specific data into messaging copy. PPF interpolates the player's own behavioural data (deposit amounts, time-on-site, limit values) into the copy at run-time. EPF builds on PPF by additionally incorporating algorithmic system-derived data (behavioural risk score, normative comparison).

At R3, the PPF/EPF distinction collapses into a single 'unified' content type. This operational simplification reflects the design intuition that high-risk-tier messaging benefits from standardised, predictable wording rather than personalised variation. The collapse is operationalised in the content_type column of the tool_variants and message_templates sheets.

6.4 Localisation

Localisation is set at the architecture level, not per-tool. UK: language en-GB, currency currency_gbp. NL: language nl-NL, currency currency_eur. Every message template in 05.1 is in en-GB; every template in 05.2 is in nl-NL. Currency placeholders interpolate to the architecture's currency at run-time. The two architectures use translation-equivalent message templates: per-tool copy intensity (mean word count, range) is substantively similar.

7. Tool exclusions

Tools considered during Stage 3 selection but not built were documented in the excluded_tools sheet of each architecture file. Each excluded variant retains its specification (the variant could be deployed if the exclusion reason were removed) and is recorded with its exclusion class and reason.

7.1 Exclusion classes

regulatory

Tool is excluded because regulation either prohibits it outright or already provides equivalent protection. NL examples: Bonus Ban and Promo Ban excluded under KOA

	2021 promotional restrictions.
operator	Tool is excluded due to operator-specific business decisions (e.g., contractual obligations, brand positioning). No current examples in either architecture.
technical	Tool is excluded because of infrastructure constraints. UK example: Automatic Withdrawal Limit (AWL) excluded because UK banking infrastructure does not support the required automatic-debit operation.
evidence	Tool is excluded because the evidence base did not support inclusion at acceptable confidence. No current examples in either architecture; this class is reserved for future architectures where additional tools may be considered and excluded on evidentiary grounds.

7.2 UK exclusions

UK architecture: 1 excluded tool (Automatic Withdrawal Limit), generating 6 excluded variants. Exclusion class: technical. Reason: UK banking infrastructure does not support the automatic-debit operation that the AWL tool requires. The variants remain specified (mandatory equivalents to the NL specification) but were not built.

7.3 NL exclusions

NL architecture: 2 excluded tools (Bonus Ban and Promo Ban), generating 12 excluded variants. Exclusion class: regulatory. Reason: KOA 2021 promotional restrictions already restrict bonuses and promotions to the operator at the regulatory level, making operator-side bans on these features unnecessary. The variants remain specified but were not built.

7.4 Cross-architecture pattern

The two exclusion patterns illustrate the four-class exclusion framework with complementary cases. The NL regulator removes tools by superseding them: the regulation already does the work the tool was designed to do, so the tool becomes redundant. The UK infrastructure removes tools by being unable to support them: the regulation permits the tool, the operator wants the tool, but the technical environment does not allow deployment. These two cases capture two of the four exclusion classes; the remaining two (operator and evidence) are reserved for cases not represented in the current architecture pair.

8. Jurisdiction-specific divergence

The UK and NL architectures share a common framework foundation, a common schema, and 12 shared tools. They diverge on tool inventory, parameter values, policy rule structure, and regulatory grounding. This section documents the substantive differences and frames them within the regulatory rigidity vs operator discretion thesis.

8.1 Tool inventory divergence

Three tools differ in deployment status across the architectures: Bonus Ban (UK only), Promo Ban (UK only), Automatic Withdrawal Limit (NL only). All three differences are driven by jurisdictional regulation, not by operator preference. The UK has Bonus Ban and Promo Ban because LCCP permits operator-side promotional restrictions as RG features. The NL excludes them because KOA 2021 already restricts operator promotional behaviour at the regulatory level. The NL has AWL because KOA 2021 mandates a wallet balance limit (saldolimiet); the UK excludes it because UK banking infrastructure does not support the implementation.

8.2 Parameter value divergence

Where the same tool is deployed in both architectures, parameter values often differ. Two parameters show the most substantive divergence:

Increase cooldown values	UK uses 6 distinct values (24h, 48h, none, disable_24h, disable_24h_daily_168h_weekly, and cannot-disable at R3); NL uniformly applies 168h across all tiers. NL cooldown rigidity is mandated by KOA 2021.
Limit-set modes	UK uses 10 distinct modes; NL uses 3. UK exposes the full design space of player-set, player-elected, and system-set modes; NL limits the design space to KOA 2021's prescribed options.

8.3 Policy rule structure divergence

UK has 78 rules across 10 sections; NL has 133 rules across 12 sections. The NL architecture has more rules and more sections because KOA 2021's regulatory text decomposes tailored limit types into distinct rule families, and because YA provisions in KOA 2021 require additional rules for players under 24.

8.4 Cross-jurisdictional framing

The cross-jurisdictional differences support a single methodological point: NL uniformity is regulatory (KOA 2021 prescribes parameter values, which propagate through the architecture as a smaller design space and a higher density of regulatory citation in policy rules); UK variability is operator-driven (LCCP permits operator design discretion, which propagates through the architecture as a larger design space and a higher density of empirical-data and operator-specification citation in policy rules).

The same framework produces two architectures; this is the methodological core of the translational design process. The framework supplies the design space; the architectures select from that design space within jurisdiction-specific constraints. The differences between UK and NL architectures are not framework-driven but constraint-driven, and the constraint sources are documented at the policy-rule level via the source taxonomy.

9. Reproducibility and audit trail

9.1 Provenance trace

Each architecture tool can be traced back to its framework provenance via 04.3 Framework to Architecture Translation. Sheet 1 of 04.3 maps each architecture tool to one or more framework tools with translation type (Direct, Selective, Consolidation, Extension, or Selective + Extension) and rationale. Sheet 2 documents the framework dimensions inherited as Architecture-level conventions.

Each framework tool can be traced forward to its architecture deployment status via 04.3 Sheet 3 (Framework Maturity Map). The Maturity Map documents, for each of the 210 framework tools, whether it is deployed in the UK architecture, the NL architecture, both, neither, or contributes to convention inheritance only.

9.2 Versioning

Each architecture file carries version metadata in its _index sheet. Variant IDs are stable across versions; once assigned, an ID is never reused or reassigned. Excluded variants retain their assigned IDs in the excluded_tools sheet rather than being removed from the ID space, preserving the audit trail of Stage 3 selection decisions.